DAT 201: Thematic Inquiry

Why Hockey Teams Should Add Tracking-Based Pre-Shot Movement Metrics to Classical xG Models

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April 17, 2026

Hockey offense is built around creating space near the net before defenders can recover. A team can do that by carrying the puck, passing quickly, cycling along the boards, or shooting before the defense gets set. Analysts often summarize those attacks with shot-based metrics such as expected goals, xGF%, RAPM, and GAR. Those measures are useful because they turn many events into a few interpretable numbers, but they mostly begin from the shot itself or from on-ice results after the fact. That creates a limit. Two shots can come from similar locations while being created in very different ways. One may come after the offense holds the puck in place and lets the defense get set. Another may come after a fast lateral pass forces the goalie and defenders to slide across the width of the rink. My thesis is that hockey teams should supplement classical xG-style metrics with tracking-based measures of pre-shot puck movement and passing context, because spatio-temporal data shows how a chance was created, not only where it ended. In simple terms, a shot from 35 feet after a static possession is not the same as a shot from 35 feet after the defense has been pulled out of shape.

What Existing Tracking Research Shows

Yu et al. (2019) make this point directly with a large Sportlogiq spatio-temporal dataset drawn from professional hockey. Their most important result for this paper is simple: at 5-on-5, high pre-shot east-west pace increased shooting percentage by 38% over low pre-shot east-west pace even though the average shot distance remained in roughly the same 37-40 foot band. Here, east-west pace mainly refers to offensive puck movement across the width of the rink before the shot. Tracking data records the puck's changing coordinates over time, so researchers can estimate how much and how quickly the puck moved laterally before the release. That movement matters because it usually forces the goalie and defenders to move laterally too. The implication is that movement before the shot changes outcomes even when shot location stays broadly similar. A model that only looks at the final shot coordinates leaves out part of the story.

Radke, Brecht, and Radke (2022) show the same idea from a passing-lane perspective using 198 NHL games of puck and player tracking data. They found that about 59% of completed one-bank passes had an indirect lane that was at least as open as the direct alternative, and their expected-movement model reduced receiver-location error in more than 94% of completed passes. In plain language, the smartest pass is often not the obvious straight-line pass that a static event log would suggest. Tracking data matters because it can measure the geometry and timing that make a play available in the first place. Overall, these sources push the animating question in a more specific direction. Instead of asking, in general, how tracking can create insights beyond xG, a sharper claim is that tracking is especially valuable when it measures pre-shot movement and passing context. Those are exactly the parts of offense that shot-only models tend to flatten.

What the Big Data Cup Sample Adds

To test that claim with my own data, I used the public 2026 Big Data Cup release, which contains synchronized event and tracking files for 10 anonymized games. The full sample includes 1132 shot attempts, of which 885 came at 5-on-5. I then linked each 5-on-5 shot to the most recent completed

play by the same team within five seconds, which produced 655 even-strength play-to-shot sequences. For each sequence, I measured pre-shot lateral puck movement as the absolute east-west change in the completed play's coordinates. I focus on whether the shot finished on net, because in a ten-game public sample that is a more stable outcome than goals alone.

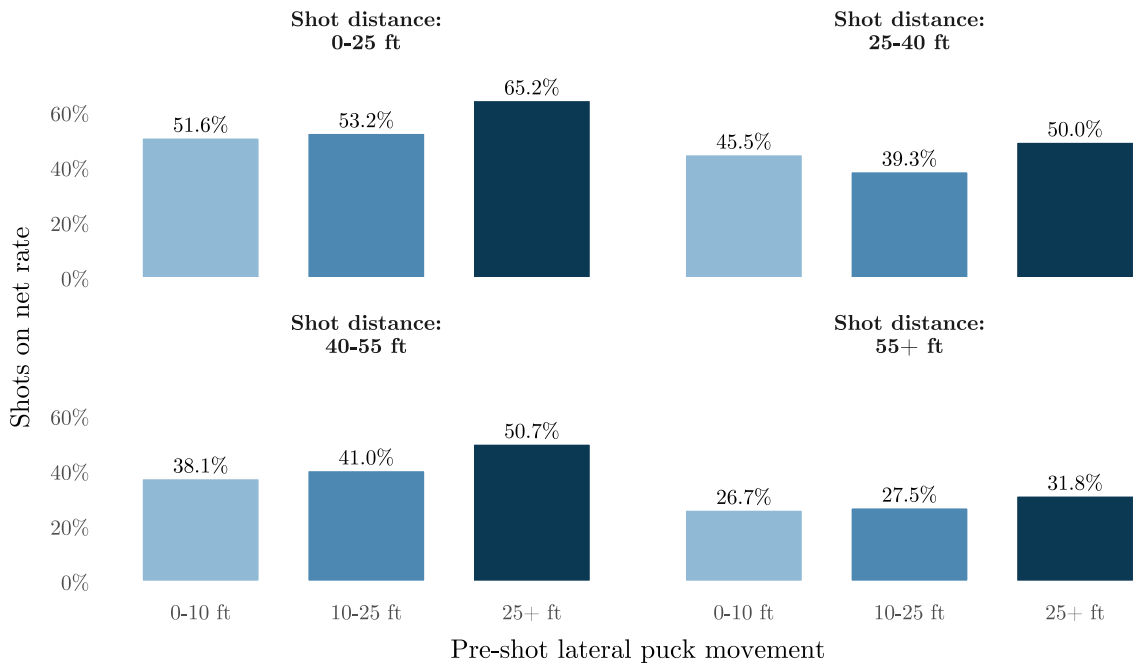

Figure 1: Within every shot-distance band, the highest on-net rate comes from plays with at least 25 feet of lateral puck movement before the shot. Facets show shot-distance groups. Sample restricted to 5-on-5 shots preceded by a completed same-team play within five seconds (n = 655).

Figure 1 supports the thesis clearly. In every shot-distance band, the highest on-net rate comes from the largest movement group, the plays with at least 25 feet of lateral puck movement. Among shots from 0-25 feet, the on-net rate rises from 51.6% after small 0-10 foot movement to 65.2% after 25+ feet of movement. In the 25-40 foot band, it rises from 45.5% to 50.0%. In the 40-55 foot band, it rises from 38.1% to 50.7%. Even on long shots from 55+ feet, the biggest east-west plays still perform better, 26.7% versus 31.8%.

Just as importantly, this is not simply a closer-shot story. A natural objection would be that the high-movement plays only look better because they end much nearer to the net. In this sample, that is not what happens: the biggest-movement group averaged 39.7 feet from the net, while the smallest-movement group averaged 39.1 feet, which is nearly identical. To make that comparison explicit, I estimated the logistic model

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1\,\mathrm{ShotDistance}_i + \beta_2\,\mathrm{AbsY}_i + \beta_3\,\mathrm{LateralMovement}_i,$$

where p_i is the probability that shot i finished on net, $\mathrm{ShotDistance}_i$ is the shot's distance from the goal, AbsY_i is its absolute side-to-side location, and $\mathrm{LateralMovement}_i$ is the pre-shot east-west puck movement measured in 10-foot units. In the fitted model, the movement coefficient is $\hat{\beta}_3 = 0.117$, while the shot-distance coefficient is $\hat{\beta}_1 = -0.030$. So, holding the other terms fixed, each additional 10 feet of lateral pre-shot movement increased the odds that a shot ended on net by about 12.5%,

whereas shots taken farther from the net became less likely to finish on net. The categorical version of the same model shows that the 25+ foot group had about 1.53 times the on-net odds of the 0-10 foot group. This does not mean lateral movement explains every good chance, but it does mean the process leading into the shot carries independent information that a shot-only model can miss.

This result fits well with the published research. Yu et al. argue that east-west pace changes shooting outcomes even when distance is held roughly constant; the Big Data Cup sample shows the same pattern in public competition data. Radke et al. show why that should not be surprising: tracking data captures passing choices and receiver movement that ordinary shot logs compress away. The shared lesson is that offensive threat is not merely about where a shot came from. It is also about whether the defense had time to get set.

Conclusion

The two sources support the same argument from different angles. Yu et al. show at larger scale that pre-shot movement changes shot success beyond simple distance. Radke et al. show that tracking can quantify passing options and moving targets that event summaries miss. My own analysis of the 2026 Big Data Cup sample extends that logic with a concrete public-data example: larger lateral puck movement before the shot is associated with better shot outcomes within every distance band I examined. That is enough evidence to defend a specific thesis: teams should not replace xG with tracking metrics, but they should add tracking-based pre-shot movement measures if they want a fuller picture of offensive creation.

A reasonable counterargument is that modern xG models already include some context, such as shot type, rush flags, or rebound indicators, and that my public sample is much smaller than a full NHL season. I do not think that is a major problem for the thesis, because I am arguing for supplementation rather than replacement. If pre-shot movement still matters after holding shot distance roughly constant, then teams lose information when they collapse everything back to the shot alone. The right response is broader validation, not a return to purely shot-based models.

These methods also suggest practical coaching questions. Offensively, coaches may want to create more sequences that move the puck laterally before the shot, especially if those sequences force the goalie to reset. Defensively, coaches may want to pressure puck carriers earlier and steer the puck toward the boards so that cross-rink movement is harder to complete.

The most useful future dataset would be frame-level optical or pose-tracking data that identifies goalie sightlines and defender stick positions at the moment of the pass and shot. That would let analysts distinguish two chances from the same x-y location when one is a clean one-timer into an open lane and the other is a screened point shot through traffic. Another natural extension would be to test whether lateral movement farther from the shot, such as 10-5 seconds beforehand, still matters or whether the effect is concentrated in the final seconds.

References

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